Functional Network Disruption as Biomarkers for Adolescent Concussion

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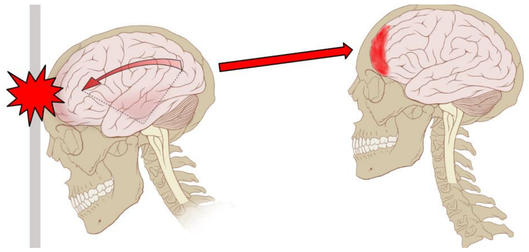
Introduction



- Concussion and trauma-related injuries affect around 400,000 Canadians annually
- Improper or late diagnosis can lead to long-term changes to cognitive and behavioural health
- Most existing diagnostic approaches over rely on physical symptoms and lack sensitivity and reproducibility
- Concussion is known to heavily disrupt the alpha frequency, the strongest signal in the brain

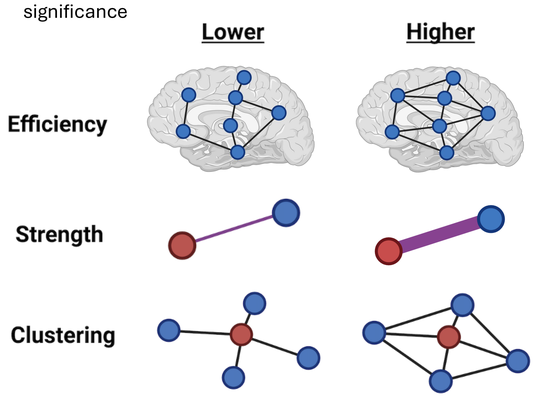
Aims and Hypothesis

- Aims: Model adolescent concussion patients' brain functional connectivity network using **graph theory** to search for changes and possible concussion biomarkers
- **Hypothesis: Brain functional network organization is disrupted in concussion compared to typical controls**

Methods

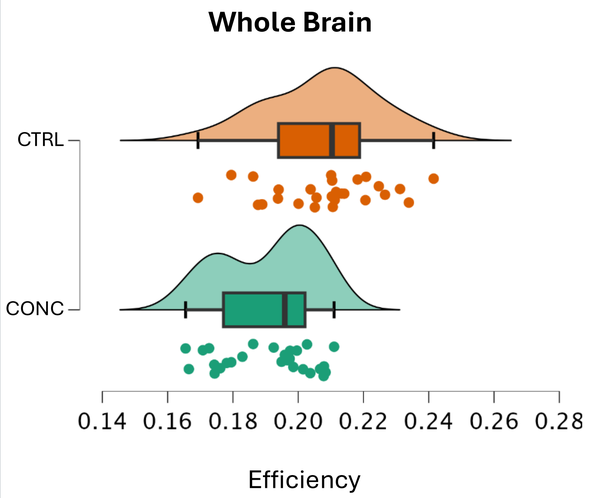
Participant characteristics	Concussion	Typical controls	Statistical tests
Sample size	27	28	-
Age (years)	14.96 ± 1.53	14.58 ± 1.30	N.S.
Sex (M/F)	8/19	9/19	N.S.
Mean time since last concussion (days)	709.3	-	-
Average number of concussions	2	-	-

- Magnetoencephalography scan in 54 regions of interest defined by the **Automated Anatomical Labelling atlas**
- Network analysis performed using **Fieldtrip** and **Brain Connectivity Toolboxes**
- Appropriate tests were used to assess group differences; **Benjamini-Hochberg** procedure used to assess significance

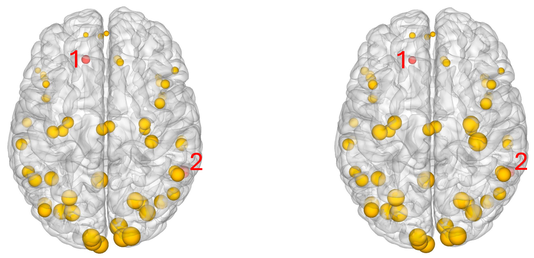


Results

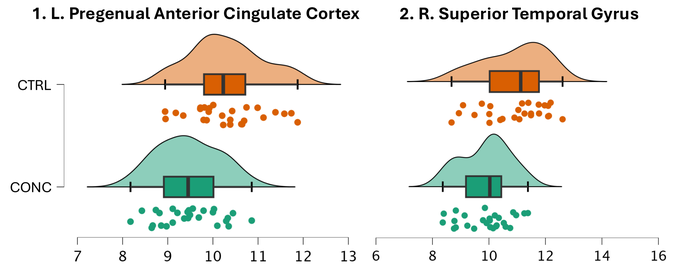
Efficiency



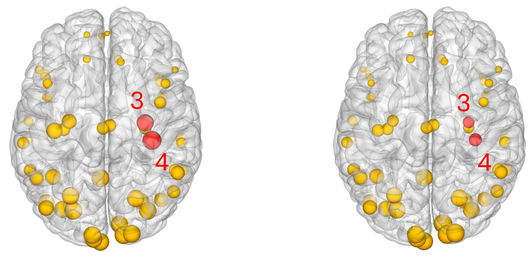
Strength



Each dot represents the averaged strength value of a brain region; red dot represents most significantly affected node



Clustering



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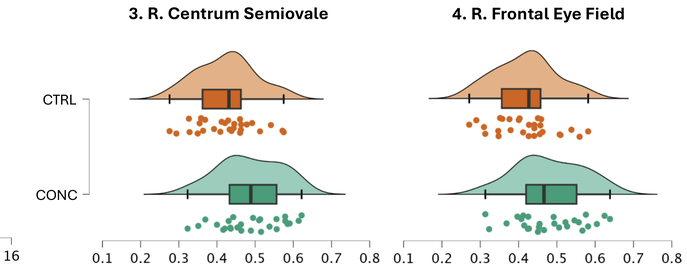


Efficiency Independent Samples T-Test

Test	Statistic	d.f.	p-value
Student	-3.946	53.00	< 0.001
Welch	-3.959	52.03	< 0.001

Strengths Independent Samples T-Test (Region 1)

Test	Statistic	d.f.	p-value
Student	-4.160	52.00	<0.001
Welch	-4.160	50.77	<0.001

Strengths Independent Samples T-Test (Region 2)

Test	Statistic	d.f.	p-value
Student	-3.706	51.00	<0.001
Welch	-3.688	46.94	<0.001

Clustering Independent Samples T-Test (Region 3)

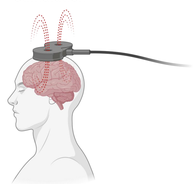
Test	Statistic	d.f.	p-value
Student	2.949	53.00	0.005
Welch	-2.946	52.51	0.005

Clustering Independent Samples T-Test (Region 4)

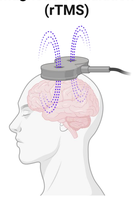
Test	Statistic	d.f.	p-value
Student	2.753	53.00	0.008
Welch	2.748	51.97	0.008

Discussion

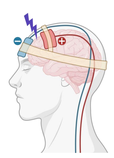
Transcranial magnetic stimulation



Rhythmic transcranial magnetic stimulation (rTMS)

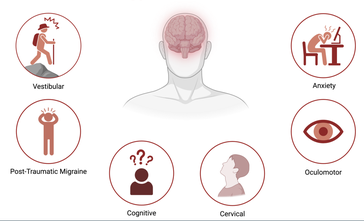


Transcranial direct current stimulation (tDCS)



- Altered network organization in concussion patients compared to controls
- Disrupted connectivity in alpha frequency is a **potential concussion biomarker**
- **Precision medicine:** affected regions could act as potential targets for neurostimulation therapies
- Mapping allows more specific diagnosis; distinguishing different subtypes
- Earlier and precise diagnoses allows for **more immediate treatment**; preventing worsening symptoms and further brain health deterioration, particularly in **vulnerable demographic such as adolescents and children**

Subtypes of Concussion



Acknowledgements

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